Brian Matejek

Computer Science · Artificial Intelligence · Data Science · Machine Learning · Algorithms

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Summary

I am a research scientist at SRI International interested in artificial intelligence, machine learning, compression, and algorithms. Some of my recent work has focused on creating high assurance machine learning solutions in healthcare and cybersecurity settings (TMLR 2025, ISBI 2025, ISBI 2024). I also study energy-constrained algorithms and artificial intelligence (NeurIPS 2025). I received my Ph.D. from Harvard University, working under the guidance of Hanspeter Pfister in the Visual Computing Group. My doctoral research concerned biologically-aware algorithms for connectomics, the interdisciplinary field between computer science and neuroscience. Specifically, I focused on the compression, error correction, and volume processing of the terabyte label volumes and graph analysis on the neuronal wiring diagrams. Before that, I completed my master's and bachelor's degrees at Princeton University working closely with Thomas Funkhouser.

Publications

2026 Aurelien Ghiglino, Daniel Elenius, Anirban Roy, Ramneet Kaur, Manoj Acharya, Colin Samplawski, **Brian Matejek**, Susmit Jha, Juan Alonso, Adam D. Cobb. "Do Diffusion Models Dream of Electric Planes?" Discrete and Continuous Simulation-Based Inference for Aircraft Design, in *International Conference on Machine Learning*.

2026 Fernando Perez-Cruz, Briland Hitaj, Giovanna Maria Dimitri, Oishi Deb, Amine M'Charrak, Shikhar Srivastava, Christopher Kanan, Anup Kumar Gupta, Zefang Liu, Tarun Kumar, Álvaro Galisteo Bermúdez, Lisa Alazraki, Vasilios Mavroudis, Yuyang Xue, **Brian Matejek**, Swati Rajwal, Avinash Kumar Pandey, Ayyüce Begüm Bektaş, Anil B Murthy, Feng Chen, Hamed Babaei Giglou, Jennifer D'Souza, Huascar Sanchez, Michael Kraus, Isamu Lautenschläger, Ziyang Wang, Xuanli He, Hyun Song Shin. The Unreasonable Benchmark, in *Data-Centric Machine Learning Research*.

2026 Alexander M. Berenbeim, Ramneet Kaur, Adam Cobb, **Brian Matejek**, Anirban Roy, Susmit Jha, Nathaniel D. Bastian. Relevance Structures are a Necessary Prerequisite for Artificial Metacognition, in *IEEE Intelligent Systems*.

2026 Mitchell Chervu Johnston, **Brian Matejek**. Overcoming Unforeseeable AI Errors: A Proposal, in *ssrn preprint ssrn:6344559*.

2025 Yi Jiang, Malyaban Bal, **Brian Matejek**, Susmit Jha, Adam Cobb, Abhronil Sengupta. Spatio-Temporal Pruning for Compressed Spiking Large Language Models, in *arXiv preprint arXiv:2508.20122*.

2025 Malyaban Bal, **Brian Matejek**, Susmit Jha, Adam D. Cobb. SpikingVTG: A Spiking Detection Transformer for Video Temporal Grounding, in *Neural Information Processing Systems (NeurIPS)*.

2025 SangHyuk Kim, Edward Gaibor, **Brian Matejek**, Daniel Haehn. Melanoma Detection with Uncertainty Quantification, in *IEEE International Symposium on Biomedical Imaging (ISBI)*.

2025 **Brian Matejek**, Ashish Gehani, Nathaniel D. Bastian, Daniel J. Clouse, Bradford J. Kline, Susmit Jha. SAFE-NID: Self-Attention with Normalizing-Flow Encodings for Network Intrusion Detection, in *Transactions on Machine Learning Research (TMLR)*.

2024 **Brian Matejek**, Daniel Elenius, Cale Gentry, David Stoker, Adam Cobb. Resource-Constrained Heuristic for Max-SAT, in *arXiv preprint arXiv:2410.09173*.

2024 Akshata Tiwari, **Brian Matejek**, Daniel Haehn. Non-Invasive Stress Monitoring from Video, in *IEEE International Symposium on Biomedical Imaging (ISBI)*.

2024 Adam Cobb, **Brian Matejek**, Daniel Elenius, Anirban Roy, Susmit Jha. Direct Amortized Likelihood Ratio Estimation, in *Thirty-Eighth AAAI Conference on Artificial Intelligence (AAAI)*.

2022 Tianyi Chen, **Brian Matejek**, Michael Mitzenmacher, and Charalampos E. Tsourakakis. Algorithmic Tools for Understanding the Motif Structure of Networks, in *Proceedings of European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD)*.

2022 **Brian Matejek**, Donglai Wei, Tianyi Chen, Charalampos E. Tsourakakis, Michael Mitzenmacher, and Hanspeter Pfister. Edge-Colored Directed Subgraph Enumeration on the Connectome, in *Nature Scientific Reports*.

2022 **Brian Matejek**[†], Tim Franzmeyer[†], Donglai Wei, Xueying Wang, Jinglin Zhao, Kálmán Palágyi, Jeff W. Lichtman, and Hanspeter Pfister. Scalable Biologically-Aware Skeleton Generation for Connectomic Volumes, in *IEEE: Transactions on Medical Imaging*.

2021 Donglai Wei, Kisuk Lee, Hanyu Li, Ran Lu, J. Alexander Bae, Zequan Liu, Lifu Zhang, Márcia dos Santos, Zudi Lin, Thomas Uram, Xueying Wang, Ignacio Arganda-Carreras, **Brian Matejek**, Narayanan Kasthuri, Jeff W. Lichtman, and Hanspeter Pfister. AxonEM Dataset: 3D Axon Instance Segmentation of Brain Cortical Regions, in *Springer: International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.

2020 Zudi Lin, Donglai Wei, Won-Dong Jang, Siyan Zhou, Xupeng Chen, Xueying Wang, Richard Schalek, Daniel Berger, **Brian Matejek**, Lee Kamentsky, Adi Peleg, Daniel Haehn, Thouis R. Jones, Toufiq Parag, Jeff Lichtman, and Hanspeter Pfister. Two Stream Active Query Suggestion for Active Learning in Connectomics, in *Proceedings of European Conference on Computer Vision (ECCV)*.

2019 **Brian Matejek**, Donglai Wei, Xueying Wang, Jinglin Zhao, Kálmán Palágyi, and Hanspeter Pfister. Synapse-Aware Skeleton Generation For Neural Circuits, in *Springer: International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.

2019 **Brian Matejek**, Daniel Haehn, Haidong Zhu, Donglai Wei, Toufiq Parag, and Hanspeter Pfister. Biologically-Constrained Graphs for Global Connectomics Reconstruction, in *Proceedings of IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

2018 Konstantin Dmitriev, Toufiq Parag, **Brian Matejek**, Arie E Kaufman, and Hanspeter Pfister. Efficient Correction for EM Connectomics with Skeletal Representation, in *British Machine Vision Conference (BMVC)*.

2018 Michael Behrisch, Dirk Streeb, Florian Stoffel, Daniel Seebacher, **Brian Matejek**, Stefan Hagen Weber, Sebastian Mittelstaedt, Hanspeter Pfister, and Daniel Keim. Commercial Visual Analytics Systems–Advances in the Big Data Analytics Field, in *IEEE Transactions on Visualization and Computer Graphics (TVCG)*.

2017 **Brian Matejek**, Daniel Haehn, Fritz Lekschas, Michael Mitzenmacher, and Hanspeter Pfister. Compresso: Efficient Compression of Segmentation Data for Connectomics, in *Springer: International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*.

2017 Daniel Haehn, John Hoffer, **Brian Matejek**, Adi Suissa-Peleg, Ali K Al-Awami, Lee Kamentsky, Felix Gonda, Eagon Meng, William Zhang, Richard Schalek, Alyssa Wilson, Toufiq Parag, Johanna Beyer, Verena Kaynig, Thouis R. Jones, James Tompkin, Markus Hadwiger, Jeff W. Lichtman, and Hanspeter Pfister. Scalable Interactive Visualization for Connectomics, in *MDPI Informatics*.

2017 Toufiq Parag, Fabian Tschopp, William Grisaitis, Srinivas C. Turaga, Xuewen Zhang, **Brian Matejek**, Lee Kamentsky, Jeff W. Lichtman, and Hanspeter Pfister. Anisotropic EM Segmentation by 3D Affinity Learning and Agglomeration, in *arXiv preprint arXiv:1707.08935*.

2015 David Dohan, **Brian Matejek**, and Thomas Funkhouser. Learning Hierarchical Semantic Segmentations of LIDAR Data, in *IEEE International Conference of 3D Vision*.

Workshops

2026 Mitchell Chervu Johnston, **Brian Matejek**. Bridging the Gap Between Tort Law and Unforeseeable AI Errors, in *Trustworthy AI For Good* at *International Conference on Machine Learning (ICML)*.

2026 Mitchell Chervu Johnston, **Brian Matejek**. Bridging the Gap Between Tort Law and Unforeseeable AI Errors, in *Workshop on Technical AI Governance* at *International Conference on Machine Learning (ICML)*.

2025 Aurelien Ghiglino, Daniel Elenius, Anirban Roy, Ramneet Kaur, Manoj Acharya, Colin Samplawski, **Brian Matejek**, Susmit Jha, Juan Alonso, Adam D. Cobb. A Preliminary Study into the Conceptual Design of Aircraft using Simulation-Based Inference, in *Machine Learning and the Physical Sciences Workshop* at *Neural Information Processing Systems (NeurIPS)*.

2025 Krishiv Agarwal, Ramneet Kaur, Colin Samplawski, Manoj Acharya, Anirban Roy, Daniel Elenius, **Brian Matejek**, Adam D. Cobb, Susmit Jha. Breaking Bad: Interpretability-Based Safety Audits of State-of-the-Art LLMs, in *Lock-LLM Workshop: Prevent Unauthorized Knowledge Use from Large Language Models* at *Neural Information Processing Systems (NeurIPS)*.

2024 Ramneet Kaur, Colin Samplawski, Adam D. Cobb, Anirban Roy, **Brian Matejek**, Manoj Acharya, Daniel Elenius, Alexander Michael Berenbeim, John A. Pavlik, Nathaniel D. Bastian, Susmit Jha. Enhancing Semantic Clustering for Uncertainty Quantification & Conformal Prediction by LLMs, in *Statistical Foundations of LLMs and Foundation Models* at *Neural Information Processing Systems (NeurIPS)*.

2024 Ramneet Kaur, Colin Samplawski, Adam D. Cobb, Anirban Roy, **Brian Matejek**, Manoj Acharya, Daniel Elenius, Alexander Michael Berenbeim, John A. Pavlik, Nathaniel D. Bastian, Susmit Jha. Addressing Uncertainty in LLMs to Enhance Reliability in Generative AI, in *Safe Generative AI* at *Neural Information Processing Systems (NeurIPS)*.

2024 Malyaban Bal, **Brian Matejek**, Susmit Jha, Adam D. Cobb. SpikingVTG: Saliency Feedback Gating Enabled Spiking Video Temporal Grounding, in *Adaptive Foundation Models Workshop* at *Neural Information Processing Systems (NeurIPS)*.

2024 Malyaban Bal, **Brian Matejek**, Susmit Jha, Adam D. Cobb. SpikingVTG: Saliency Feedback Gating Enabled Spiking Video Temporal Grounding, in *Machine Learning and Compression Workshop* at *Neural Information Processing Systems (NeurIPS)*.

2024 Akshata Tiwari, **Brian Matejek**, Daniel Haehn. Non-Invasive Stress Monitoring from Video, in *Women in Machine Learning Workshop* at *Neural Information Processing Systems (NeurIPS)*.

2024 **Brian Matejek**, Ashish Gehani, Nathaniel D. Bastian, Daniel J. Clouse, Bradford J. Kline, Susmit Jha. Safeguarding Network Intrusion Detection Models from Zero-day Attacks and Concept Drift, in *Artificial Intelligence for Cyber Security (AICS) Workshop* at *AAAI Conference on Artificial Intelligence (AAAI)*.

2023 Adam D. Cobb, **Brian Matejek**, Daniel Elenius, Anirban Roy, and Susmit Jha. Direct Amortized Likelihood Ratio Estimation, in *Machine Learning and the Physical Sciences Workshop* at *Neural Information Processing Systems (NeurIPS)*.

Invited Talks, Oral Exams, and Defenses

2024 Safeguarding Network Intrusion Detection Models from Zero-day Attacks and Concept Drift, Artificial Intelligence for Cybsersecurity Workshop — *AICS at AAAI*

2023	Introduction to Generative AI, Tech Workshop at Micro Conference	*THRIVE-WiSE*
2023	High-Assurance Machine Learning for Cybersecurity, Guest Lecture for CS410: Introduction to Software Engineering	*UMass Boston*
2021	Biologically-Aware Algorithms for Connectomics, Dissertation Defense	*Harvard University*
2019	Efficient Error Correction for Connectomics, Bioimage Computing Workshop	*CVPR*
2019	Synapse-Aware Skeleton Generation for Neural Circuits, Connectomics Conference	*Max Planck Institute*
2018	Segmentation of Electron Micrscopy Images in Connectomics, Qualifying Exam	*Harvard University*
2016	Learning Global Features for Neuron Reconstruction in EM Images, Master's Thesis Defense	*Princeton University*

Education

Harvard University *Cambridge, MA 02138*

Ph.D. in Computer Science *Aug 2016 - May 2021*

- Thesis: Biologically-Aware Algorithms for Connectomics
- Advisor: Hanspeter Pfister
- Committee Members: Michael Mitzenmacher, Todd Zickler
- GPA: 4.00 / 4.00

Princeton University *Princeton, NJ 08544*

M.S.E. in Computer Science *Sep 2014 - May 2016*

- Thesis: Learning Global Features for Neuron Reconstruction in EM Images
- Advisor: Thomas Funkhouser
- GPA: 3.85 / 4.00

B.S.E. in Computer Science *Sep 2010 - Jun 2014*

- Independent Research: Detecting Objects Using Google Street View Data
- Independent Research: A Computational Analysis of Arbitrage Opportunities in Sports Gambling
- GPA: 3.79 / 4.00, High Honors

Honors & Scholarships

Harvard University

2017-18	**Smith Family Fellowship,** Graduate Stipend Award
2016-17	**Tess Denny Chen Graduate Student Research Fellowship,** Graduate Stipend Award

Princeton University

2014 -	**Sigma Xi,** Admitted to Scientific Research Society for Undergraduates
2012 -	**Tau Beta Pi,** Admitted to Engineering Honor Society, Top 12% of Class

Teaching Experience

Harvard University

Fall 2018	Computer Science 109A: Introduction to Data Science

Princeton University

Spr 2016	Computer Science 423: Theory of Algorithms
Fall 2015	Computer Science 402: Artificial Intelligence
Spr 2015	Computer Science 340: Reasoning About Computation
Fall 2014	Computer Science 429: Computer Vision

Mentored Students

2024	SangHyuk Kim, University of Massachusetts Boston
2021-24	Akshata Tiwari, Massachusetts Institute of Technology
2021-24	Prateeti Saran, University of Cambridge
2020-21	Simon Warchol, Harvard University
2020-21	Ke Li, Chinese Academy of Sciences
2019-20	Tim Franzmeyer, Eidgenössische Technische Hochschule Zürich (ETH Zurich)
2019	Ian Svetkey, Harvard University

2019 Antoine Alleon, École Polytechnique Fédérale de Lausanne (EPFL)
2019 Romil Sirohi, Harvard University
2019 Bruno Mlodozeniec, University of Cambridge

Professional Services

Reviewer

2026 IEEE Transactions on Network and Service Management
2026 Association for the Advancement of Artificial Intelligence (AAAI)
2024 Association for the Advancement of Artificial Intelligence (AAAI)
2022 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)
2021 IEEE Visualization Conference
2021 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)
2021 PLOS Computational Biology
2020 ISCB Conference on Intelligent Systems for Molecular Biology (ISMB)
2020 IEEE European Conference on Computer Vision (ECCV)
2019 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)

Skills

Programming Python, C++, C, LaTeX, PyTorch, Huggingface, OpenHands, Cython, Numpy, Pandas, Django, Sass, jQuery, Linux, SLURM
Clearance Top Secret Clearance